

TABLE 1 Normal Curve Areas

The entries in the body of the table correspond to the area shaded under the normal curve.

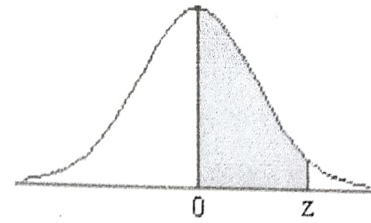
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Table 4 Values of  $t_{\alpha}^*$ 

| $\nu$ | $\alpha = 0.10$ | $\alpha = 0.05$ | $\alpha = 0.025$ | $\alpha = 0.01$ | $\alpha = 0.005$ | $\nu$ |
|-------|-----------------|-----------------|------------------|-----------------|------------------|-------|
| 1     | 3.078           | 6.314           | 12.706           | 31.821          | 63.657           | 1     |
| 2     | 1.886           | 2.920           | 4.303            | 6.965           | 9.925            | 2     |
| 3     | 1.638           | 2.353           | 3.182            | 4.541           | 5.841            | 3     |
| 4     | 1.533           | 2.132           | 2.776            | 3.747           | 4.604            | 4     |
| 5     | 1.476           | 2.015           | 2.571            | 3.365           | 4.032            | 5     |
| 6     | 1.440           | 1.943           | 2.447            | 3.143           | 3.707            | 6     |
| 7     | 1.415           | 1.895           | 2.365            | 2.998           | 3.499            | 7     |
| 8     | 1.397           | 1.860           | 2.306            | 2.896           | 3.355            | 8     |
| 9     | 1.383           | 1.833           | 2.262            | 2.821           | 3.250            | 9     |
| 10    | 1.372           | 1.812           | 2.228            | 2.764           | 3.169            | 10    |
| 11    | 1.363           | 1.796           | 2.201            | 2.718           | 3.106            | 11    |
| 12    | 1.356           | 1.782           | 2.179            | 2.681           | 3.055            | 12    |
| 13    | 1.350           | 1.771           | 2.160            | 2.650           | 3.012            | 13    |
| 14    | 1.345           | 1.761           | 2.145            | 2.624           | 2.977            | 14    |
| 15    | 1.341           | 1.753           | 2.131            | 2.602           | 2.947            | 15    |
| 16    | 1.337           | 1.746           | 2.120            | 2.583           | 2.921            | 16    |
| 17    | 1.333           | 1.740           | 2.110            | 2.567           | 2.898            | 17    |
| 18    | 1.330           | 1.734           | 2.101            | 2.552           | 2.878            | 18    |
| 19    | 1.328           | 1.729           | 2.093            | 2.539           | 2.861            | 19    |
| 20    | 1.325           | 1.725           | 2.086            | 2.528           | 2.845            | 20    |
| 21    | 1.323           | 1.721           | 2.080            | 2.518           | 2.831            | 21    |
| 22    | 1.321           | 1.717           | 2.074            | 2.508           | 2.819            | 22    |
| 23    | 1.319           | 1.714           | 2.069            | 2.500           | 2.807            | 23    |
| 24    | 1.318           | 1.711           | 2.064            | 2.492           | 2.797            | 24    |
| 25    | 1.316           | 1.708           | 2.060            | 2.485           | 2.787            | 25    |
| 26    | 1.315           | 1.706           | 2.056            | 2.479           | 2.779            | 26    |
| 27    | 1.314           | 1.703           | 2.052            | 2.473           | 2.771            | 27    |
| 28    | 1.313           | 1.701           | 2.048            | 2.467           | 2.763            | 28    |
| 29    | 1.311           | 1.699           | 2.045            | 2.462           | 2.756            | 29    |
| inf.  | 1.282           | 1.645           | 1.960            | 2.326           | 2.576            | inf.  |



**Table 5**    Values of  $\chi^2_{\alpha}$ \*

| $\nu \downarrow \alpha \uparrow$ | $\alpha = 0.995$ | $\alpha = 0.99$ | $\alpha = 0.975$ | $\alpha = 0.95$ | $\alpha = 0.05$ | $\alpha = 0.025$ | $\alpha = 0.01$ | $\alpha = 0.005$ |
|----------------------------------|------------------|-----------------|------------------|-----------------|-----------------|------------------|-----------------|------------------|
| 1                                | 0.0000393        | 0.000157        | 0.000982         | 0.00393         | 3.841           | 5.024            | 6.635           | 7.879            |
| 2                                | 0.0100           | 0.0201          | 0.0506           | 0.103           | 5.991           | 7.378            | 9.210           | 10.597           |
| 3                                | 0.0717           | 0.115           | 0.216            | 0.352           | 7.815           | 9.348            | 11.345          | 12.838           |
| 4                                | 0.207            | 0.297           | 0.484            | 0.711           | 9.488           | 11.143           | 13.277          | 14.860           |
| 5                                | 0.412            | 0.554           | 0.831            | 1.145           | 11.070          | 12.832           | 15.086          | 16.750           |
| 6                                | 0.676            | 0.872           | 1.237            | 1.635           | 12.592          | 14.449           | 16.812          | 18.548           |
| 7                                | 0.989            | 1.239           | 1.690            | 2.167           | 14.067          | 16.013           | 18.475          | 20.278           |
| 8                                | 1.344            | 1.646           | 2.180            | 2.733           | 15.507          | 17.535           | 20.090          | 21.955           |
| 9                                | 1.735            | 2.088           | 2.700            | 3.325           | 16.919          | 19.023           | 21.666          | 23.589           |
| 10                               | 2.156            | 2.558           | 3.247            | 3.940           | 18.307          | 20.483           | 23.209          | 25.188           |
| 11                               | 2.603            | 3.053           | 3.816            | 4.575           | 19.675          | 21.920           | 24.725          | 26.757           |
| 12                               | 3.074            | 3.571           | 4.404            | 5.226           | 21.026          | 23.337           | 26.217          | 28.300           |
| 13                               | 3.565            | 4.107           | 5.009            | 5.892           | 22.362          | 24.736           | 27.688          | 29.819           |
| 14                               | 4.075            | 4.660           | 5.629            | 6.571           | 23.685          | 26.119           | 29.141          | 31.319           |
| 15                               | 4.601            | 5.229           | 6.262            | 7.261           | 24.996          | 27.488           | 30.578          | 32.801           |
| 16                               | 5.142            | 5.812           | 6.908            | 7.962           | 26.296          | 28.845           | 32.000          | 34.267           |
| 17                               | 5.697            | 6.408           | 7.564            | 8.672           | 27.587          | 30.191           | 33.409          | 35.718           |
| 18                               | 6.265            | 7.015           | 8.231            | 9.390           | 28.869          | 31.526           | 34.805          | 37.156           |
| 19                               | 6.844            | 7.633           | 8.907            | 10.117          | 30.144          | 32.852           | 36.191          | 38.582           |
| 20                               | 7.434            | 8.260           | 9.591            | 10.851          | 31.410          | 34.170           | 37.566          | 39.997           |
| 21                               | 8.034            | 8.897           | 10.283           | 11.591          | 32.671          | 35.479           | 38.932          | 41.401           |
| 22                               | 8.643            | 9.542           | 10.982           | 12.338          | 33.924          | 36.781           | 40.289          | 42.796           |
| 23                               | 9.260            | 10.196          | 11.689           | 13.091          | 35.172          | 38.076           | 41.638          | 44.181           |
| 24                               | 9.886            | 10.856          | 12.401           | 13.884          | 36.415          | 39.364           | 42.980          | 45.558           |
| 25                               | 10.520           | 11.524          | 13.120           | 14.611          | 37.652          | 40.646           | 44.314          | 46.928           |
| 26                               | 11.160           | 12.198          | 13.844           | 15.379          | 38.885          | 41.923           | 45.642          | 48.290           |
| 27                               | 11.808           | 12.879          | 14.573           | 16.151          | 40.113          | 43.194           | 46.963          | 49.645           |
| 28                               | 12.461           | 13.565          | 15.308           | 16.928          | 41.337          | 44.461           | 48.278          | 50.993           |
| 29                               | 13.121           | 14.256          | 16.047           | 17.708          | 42.557          | 45.772           | 49.588          | 52.336           |
| 30                               | 13.787           | 14.953          | 16.791           | 18.493          | 43.773          | 46.979           | 50.892          | 53.672           |
| 40                               | 20.706           | 22.164          | 24.433           | 26.509          | 55.758          | 59.342           | 63.691          | 66.766           |
| 50                               | 27.991           | 29.707          | 32.357           | 34.764          | 67.505          | 71.420           | 76.154          | 79.490           |
| 60                               | 35.535           | 37.485          | 40.482           | 43.118          | 79.082          | 83.298           | 88.379          | 91.952           |
| 70                               | 43.275           | 45.442          | 48.758           | 51.739          | 90.531          | 95.023           | 100.425         | 104.215          |
| 80                               | 51.172           | 53.540          | 57.153           | 60.391          | 101.879         | 106.629          | 112.329         | 116.321          |
| 90                               | 59.196           | 61.754          | 65.646           | 69.126          | 113.145         | 118.136          | 124.116         | 128.299          |
| 100                              | 67.328           | 70.065          | 74.222           | 77.929          | 124.342         | 129.561          | 135.807         | 140.169          |

Table 6 (a) Values of  $F_{0.05}^*$ 

| $v_2$ = degrees of freedom for denominator | $v_1$ = Degrees of freedom for numerator |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |          |  |
|--------------------------------------------|------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|--|
|                                            | 1                                        | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    | 12    | 15    | 20    | 24    | 30    | 40    | 60    | 120   | $\infty$ |  |
| 1                                          | 161                                      | 200   | 216   | 225   | 230   | 234   | 237   | 239   | 241   | 242   | 244   | 246   | 248   | 249   | 250   | 251   | 252   | 253   | 254      |  |
| 2                                          | 18.50                                    | 19.00 | 19.20 | 19.20 | 19.30 | 19.30 | 19.40 | 19.40 | 19.40 | 19.40 | 19.40 | 19.40 | 19.40 | 19.50 | 19.50 | 19.50 | 19.50 | 19.50 | 19.50    |  |
| 3                                          | 10.10                                    | 9.55  | 9.28  | 9.12  | 9.01  | 8.94  | 8.89  | 8.85  | 8.81  | 8.79  | 8.74  | 8.70  | 8.66  | 8.64  | 8.62  | 8.59  | 8.57  | 8.55  | 8.53     |  |
| 4                                          | 7.71                                     | 6.94  | 6.59  | 6.39  | 6.26  | 6.16  | 6.09  | 6.04  | 6.00  | 5.96  | 5.91  | 5.86  | 5.80  | 5.77  | 5.75  | 5.72  | 5.69  | 5.66  | 5.63     |  |
| 5                                          | 6.61                                     | 5.79  | 5.41  | 5.19  | 5.05  | 4.95  | 4.88  | 4.82  | 4.77  | 4.74  | 4.68  | 4.62  | 4.56  | 4.53  | 4.50  | 4.46  | 4.43  | 4.40  | 4.37     |  |
| 6                                          | 5.99                                     | 5.14  | 4.76  | 4.53  | 4.39  | 4.28  | 4.21  | 4.15  | 4.10  | 4.06  | 4.00  | 3.94  | 3.87  | 3.84  | 3.81  | 3.77  | 3.74  | 3.70  | 3.67     |  |
| 7                                          | 5.59                                     | 4.74  | 4.35  | 4.12  | 3.97  | 3.87  | 3.79  | 3.73  | 3.68  | 3.64  | 3.57  | 3.51  | 3.44  | 3.41  | 3.38  | 3.34  | 3.30  | 3.27  | 3.23     |  |
| 8                                          | 5.32                                     | 4.46  | 4.07  | 3.84  | 3.69  | 3.58  | 3.50  | 3.44  | 3.39  | 3.35  | 3.28  | 3.22  | 3.15  | 3.12  | 3.08  | 3.04  | 3.01  | 2.97  | 2.93     |  |
| 9                                          | 5.12                                     | 4.26  | 3.86  | 3.63  | 3.48  | 3.37  | 3.29  | 3.23  | 3.18  | 3.14  | 3.07  | 3.01  | 2.94  | 2.90  | 2.86  | 2.83  | 2.79  | 2.75  | 2.71     |  |
| 10                                         | 4.96                                     | 4.10  | 3.71  | 3.48  | 3.33  | 3.22  | 3.14  | 3.07  | 3.02  | 2.98  | 2.91  | 2.85  | 2.77  | 2.74  | 2.70  | 2.66  | 2.62  | 2.58  | 2.54     |  |
| 11                                         | 4.84                                     | 3.98  | 3.59  | 3.36  | 3.20  | 3.09  | 3.01  | 2.95  | 2.90  | 2.85  | 2.79  | 2.72  | 2.65  | 2.61  | 2.57  | 2.53  | 2.49  | 2.45  | 2.40     |  |
| 12                                         | 4.75                                     | 3.89  | 3.49  | 3.26  | 3.11  | 3.00  | 2.91  | 2.85  | 2.80  | 2.75  | 2.69  | 2.62  | 2.54  | 2.51  | 2.47  | 2.38  | 2.38  | 2.30  | 2.30     |  |
| 13                                         | 4.67                                     | 3.81  | 3.41  | 3.18  | 3.03  | 2.92  | 2.83  | 2.77  | 2.71  | 2.67  | 2.60  | 2.53  | 2.46  | 2.42  | 2.38  | 2.34  | 2.30  | 2.25  | 2.21     |  |
| 14                                         | 4.60                                     | 3.74  | 3.34  | 3.11  | 2.96  | 2.85  | 2.76  | 2.70  | 2.65  | 2.60  | 2.53  | 2.46  | 2.39  | 2.35  | 2.31  | 2.27  | 2.22  | 2.18  | 2.13     |  |
| 15                                         | 4.54                                     | 3.68  | 3.29  | 3.06  | 2.90  | 2.79  | 2.71  | 2.64  | 2.59  | 2.54  | 2.48  | 2.40  | 2.33  | 2.29  | 2.25  | 2.20  | 2.16  | 2.11  | 2.07     |  |
| 16                                         | 4.49                                     | 3.63  | 3.24  | 3.01  | 2.85  | 2.74  | 2.66  | 2.59  | 2.54  | 2.49  | 2.42  | 2.35  | 2.28  | 2.24  | 2.19  | 2.15  | 2.11  | 2.06  | 2.01     |  |
| 17                                         | 3.45                                     | 3.59  | 3.20  | 2.96  | 2.81  | 2.70  | 2.61  | 2.55  | 2.49  | 2.45  | 2.38  | 2.31  | 2.23  | 2.19  | 2.15  | 2.10  | 2.06  | 2.01  | 1.96     |  |
| 18                                         | 4.41                                     | 3.55  | 3.16  | 2.93  | 2.77  | 2.66  | 2.58  | 2.51  | 2.46  | 2.41  | 2.34  | 2.27  | 2.19  | 2.15  | 2.11  | 2.06  | 2.02  | 1.97  | 1.93     |  |
| 19                                         | 4.38                                     | 3.52  | 3.13  | 2.90  | 2.74  | 2.63  | 2.54  | 2.48  | 2.42  | 2.38  | 2.31  | 2.23  | 2.16  | 2.11  | 2.07  | 2.03  | 1.98  | 1.93  | 1.88     |  |
| 20                                         | 4.35                                     | 3.49  | 3.10  | 2.87  | 2.71  | 2.60  | 2.51  | 2.45  | 2.39  | 2.35  | 2.28  | 2.20  | 2.12  | 2.08  | 2.04  | 1.99  | 1.95  | 1.90  | 1.84     |  |
| 21                                         | 4.32                                     | 3.47  | 3.07  | 2.84  | 2.68  | 2.57  | 2.49  | 2.42  | 2.37  | 2.32  | 2.25  | 2.18  | 2.10  | 2.05  | 2.01  | 1.96  | 1.92  | 1.87  | 1.81     |  |
| 22                                         | 4.30                                     | 3.44  | 3.05  | 2.82  | 2.66  | 2.55  | 2.46  | 2.40  | 2.34  | 2.30  | 2.23  | 2.15  | 2.07  | 2.03  | 1.98  | 1.94  | 1.89  | 1.84  | 1.78     |  |
| 23                                         | 4.28                                     | 3.42  | 3.03  | 2.80  | 2.64  | 2.53  | 2.44  | 2.37  | 2.32  | 2.27  | 2.20  | 2.13  | 2.05  | 2.01  | 1.96  | 1.91  | 1.86  | 1.81  | 1.76     |  |
| 24                                         | 4.26                                     | 3.40  | 3.01  | 2.78  | 2.62  | 2.51  | 2.42  | 2.36  | 2.30  | 2.25  | 2.18  | 2.11  | 2.03  | 1.98  | 1.94  | 1.89  | 1.84  | 1.79  | 1.73     |  |
| 25                                         | 4.24                                     | 3.39  | 2.99  | 2.76  | 2.60  | 2.49  | 2.40  | 2.34  | 2.28  | 2.24  | 2.16  | 2.09  | 2.01  | 1.96  | 1.92  | 1.87  | 1.82  | 1.77  | 1.71     |  |
| 30                                         | 4.17                                     | 3.32  | 2.92  | 2.69  | 2.53  | 2.42  | 2.33  | 2.27  | 2.21  | 2.16  | 2.09  | 2.01  | 1.93  | 1.89  | 1.84  | 1.79  | 1.74  | 1.68  | 1.62     |  |
| 40                                         | 4.08                                     | 3.23  | 2.84  | 2.61  | 2.45  | 2.34  | 2.25  | 2.18  | 2.12  | 2.08  | 2.00  | 1.92  | 1.84  | 1.79  | 1.74  | 1.69  | 1.64  | 1.58  | 1.51     |  |
| 60                                         | 4.00                                     | 3.15  | 2.76  | 2.53  | 2.37  | 2.25  | 2.17  | 2.10  | 2.04  | 1.99  | 1.92  | 1.84  | 1.75  | 1.70  | 1.65  | 1.59  | 1.53  | 1.47  | 1.39     |  |
| 120                                        | 3.92                                     | 3.07  | 2.68  | 2.45  | 2.29  | 2.18  | 2.09  | 2.02  | 1.96  | 1.91  | 1.83  | 1.75  | 1.66  | 1.61  | 1.55  | 1.50  | 1.43  | 1.35  | 1.25     |  |
| $\infty$                                   | 3.84                                     | 3.00  | 2.60  | 2.37  | 2.21  | 2.10  | 2.01  | 1.94  | 1.88  | 1.83  | 1.75  | 1.67  | 1.57  | 1.52  | 1.46  | 1.39  | 1.32  | 1.22  | 1.00     |  |



Table 6 (b) Tabic Values of  $F_{0.01}^*$

| $v_2$ = degrees of<br>freedom for<br>denominator | $v_1$ = Degrees of freedom for numerator |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |          |
|--------------------------------------------------|------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|
|                                                  | 1                                        | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    | 12    | 15    | 20    | 24    | 30    | 40    | 60    | 120   | $\infty$ |
| 1                                                | 4.052                                    | 5.000 | 5.403 | 5.625 | 5.764 | 5.859 | 5.928 | 5.982 | 6.023 | 6.056 | 6.106 | 6.157 | 6.209 | 6.235 | 6.261 | 6.287 | 6.313 | 6.339 | 6.366    |
| 2                                                | 98.50                                    | 99.00 | 99.20 | 99.20 | 99.30 | 99.30 | 99.40 | 99.40 | 99.40 | 99.40 | 99.40 | 99.40 | 99.40 | 99.50 | 99.50 | 99.50 | 99.50 | 99.50 | 99.50    |
| 3                                                | 34.10                                    | 30.80 | 29.50 | 28.70 | 28.20 | 27.90 | 27.70 | 27.50 | 27.30 | 27.20 | 27.10 | 26.90 | 26.70 | 26.60 | 26.50 | 26.40 | 26.30 | 26.20 | 26.10    |
| 4                                                | 21.20                                    | 18.00 | 16.70 | 16.00 | 15.50 | 15.20 | 15.00 | 14.80 | 14.70 | 14.50 | 14.50 | 14.20 | 14.00 | 13.90 | 13.80 | 13.70 | 13.70 | 13.60 | 13.50    |
| 5                                                | 16.30                                    | 13.30 | 12.10 | 11.40 | 11.00 | 10.70 | 10.50 | 10.30 | 10.20 | 10.10 | 9.89  | 9.72  | 9.55  | 9.47  | 9.38  | 9.29  | 9.20  | 9.11  | 9.02     |
| 6                                                | 13.70                                    | 10.90 | 9.78  | 9.15  | 8.75  | 8.47  | 8.26  | 8.10  | 7.98  | 7.87  | 7.72  | 7.56  | 7.40  | 7.31  | 7.23  | 7.14  | 7.06  | 6.97  | 6.88     |
| 7                                                | 12.20                                    | 9.55  | 8.45  | 7.85  | 7.46  | 7.19  | 6.99  | 6.84  | 6.72  | 6.62  | 6.47  | 6.31  | 6.16  | 6.07  | 5.99  | 5.91  | 5.82  | 5.74  | 5.65     |
| 8                                                | 11.30                                    | 8.65  | 7.59  | 7.04  | 6.63  | 6.37  | 6.18  | 6.03  | 5.91  | 5.81  | 5.67  | 5.52  | 5.36  | 5.28  | 5.20  | 5.12  | 5.03  | 4.95  | 4.83     |
| 9                                                | 10.60                                    | 8.02  | 6.99  | 6.42  | 6.06  | 5.80  | 5.61  | 5.47  | 5.35  | 5.26  | 5.11  | 4.96  | 4.81  | 4.73  | 4.65  | 4.57  | 4.48  | 4.40  | 4.31     |
| 10                                               | 10.00                                    | 7.56  | 6.55  | 5.99  | 5.64  | 5.39  | 5.20  | 5.06  | 4.94  | 4.85  | 4.71  | 4.56  | 4.41  | 4.33  | 4.25  | 4.17  | 4.08  | 4.00  | 3.91     |
| 11                                               | 9.65                                     | 7.21  | 6.22  | 5.67  | 5.32  | 5.07  | 4.89  | 4.74  | 4.63  | 4.54  | 4.40  | 4.25  | 4.10  | 4.02  | 3.94  | 3.86  | 3.78  | 3.69  | 3.60     |
| 12                                               | 9.33                                     | 6.93  | 5.95  | 5.41  | 5.06  | 4.82  | 4.64  | 4.50  | 4.39  | 4.30  | 4.16  | 4.01  | 3.86  | 3.78  | 3.70  | 3.62  | 3.54  | 3.45  | 3.36     |
| 13                                               | 9.07                                     | 6.70  | 5.74  | 5.21  | 4.86  | 4.62  | 4.44  | 4.30  | 4.19  | 4.10  | 3.96  | 3.82  | 3.66  | 3.59  | 3.51  | 3.43  | 3.34  | 3.25  | 3.17     |
| 14                                               | 8.86                                     | 6.51  | 5.56  | 5.04  | 4.70  | 4.46  | 4.28  | 4.14  | 4.03  | 3.94  | 3.80  | 3.66  | 3.51  | 3.43  | 3.35  | 3.27  | 3.18  | 3.09  | 3.00     |
| 15                                               | 8.68                                     | 6.36  | 5.42  | 4.89  | 4.56  | 4.32  | 4.14  | 4.00  | 3.89  | 3.80  | 3.67  | 3.52  | 3.37  | 3.29  | 3.21  | 3.13  | 3.05  | 2.96  | 2.87     |
| 16                                               | 8.53                                     | 6.23  | 5.29  | 4.77  | 4.44  | 4.20  | 4.03  | 3.89  | 3.78  | 3.69  | 3.55  | 3.41  | 3.26  | 3.18  | 3.10  | 3.02  | 2.93  | 2.84  | 2.75     |
| 17                                               | 8.40                                     | 6.11  | 5.19  | 4.67  | 4.34  | 4.10  | 3.93  | 3.79  | 3.68  | 3.59  | 3.46  | 3.31  | 3.16  | 3.08  | 3.00  | 2.92  | 2.83  | 2.75  | 2.65     |
| 18                                               | 8.29                                     | 6.01  | 5.09  | 4.58  | 4.25  | 4.01  | 3.84  | 3.71  | 3.60  | 3.51  | 3.37  | 3.23  | 3.08  | 3.00  | 2.92  | 2.84  | 2.76  | 2.67  | 2.58     |
| 19                                               | 8.19                                     | 5.93  | 5.01  | 4.50  | 4.17  | 3.94  | 3.77  | 3.63  | 3.52  | 3.43  | 3.30  | 3.15  | 3.00  | 2.92  | 2.84  | 2.76  | 2.67  | 2.58  | 2.49     |
| 20                                               | 8.10                                     | 5.85  | 4.94  | 4.43  | 4.10  | 3.87  | 3.70  | 3.56  | 3.46  | 3.37  | 3.23  | 3.09  | 2.94  | 2.86  | 2.78  | 2.69  | 2.61  | 2.52  | 2.42     |
| 21                                               | 8.02                                     | 5.78  | 4.87  | 4.37  | 4.04  | 3.81  | 3.64  | 3.51  | 3.40  | 3.31  | 3.17  | 3.03  | 2.88  | 2.80  | 2.72  | 2.64  | 2.55  | 2.46  | 2.36     |
| 22                                               | 7.95                                     | 5.72  | 4.82  | 4.31  | 3.99  | 3.76  | 3.59  | 3.45  | 3.35  | 3.26  | 3.12  | 2.98  | 2.83  | 2.75  | 2.67  | 2.58  | 2.50  | 2.40  | 2.31     |
| 23                                               | 7.88                                     | 5.66  | 4.76  | 4.26  | 3.94  | 3.71  | 3.54  | 3.41  | 3.30  | 3.21  | 3.07  | 2.93  | 2.78  | 2.70  | 2.62  | 2.54  | 2.45  | 2.35  | 2.26     |
| 24                                               | 7.82                                     | 5.61  | 4.72  | 4.22  | 3.90  | 3.67  | 3.50  | 3.36  | 3.26  | 3.17  | 3.03  | 2.89  | 2.74  | 2.66  | 2.58  | 2.49  | 2.40  | 2.31  | 2.21     |
| 25                                               | 7.77                                     | 5.57  | 4.68  | 4.18  | 3.86  | 3.63  | 3.46  | 3.32  | 3.22  | 3.13  | 2.99  | 2.85  | 2.70  | 2.62  | 2.53  | 2.45  | 2.36  | 2.27  | 2.17     |
| 30                                               | 7.56                                     | 5.39  | 4.51  | 4.02  | 3.70  | 3.47  | 3.30  | 3.17  | 3.07  | 2.98  | 2.84  | 2.70  | 2.55  | 2.47  | 2.39  | 2.30  | 2.21  | 2.11  | 2.01     |
| 40                                               | 7.31                                     | 5.18  | 4.31  | 3.83  | 3.51  | 3.29  | 3.12  | 2.99  | 2.89  | 2.80  | 2.66  | 2.52  | 2.37  | 2.29  | 2.20  | 2.11  | 2.02  | 1.92  | 1.80     |
| 60                                               | 7.08                                     | 4.98  | 4.13  | 3.65  | 3.34  | 3.12  | 2.95  | 2.82  | 2.72  | 2.63  | 2.50  | 2.35  | 2.20  | 2.12  | 2.03  | 1.94  | 1.84  | 1.73  | 1.60     |
| 120                                              | 6.85                                     | 4.79  | 3.95  | 3.48  | 3.17  | 2.96  | 2.79  | 2.66  | 2.56  | 2.47  | 2.34  | 2.19  | 2.03  | 1.95  | 1.86  | 1.76  | 1.66  | 1.53  | 1.38     |
| $\infty$                                         | 6.63                                     | 4.61  | 3.78  | 3.32  | 3.02  | 2.80  | 2.64  | 2.51  | 2.41  | 2.32  | 2.18  | 2.04  | 1.88  | 1.79  | 1.70  | 1.59  | 1.47  | 1.32  | 1.00     |